Experiment 1(d): 8-Bit Multiplication Using 8085 Microprocessor

AIM:

To write an assembly language program to multiply two 8-bit numbers stored in memory locations 4200H and 4201H and store the product in 4202H and 4203H.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

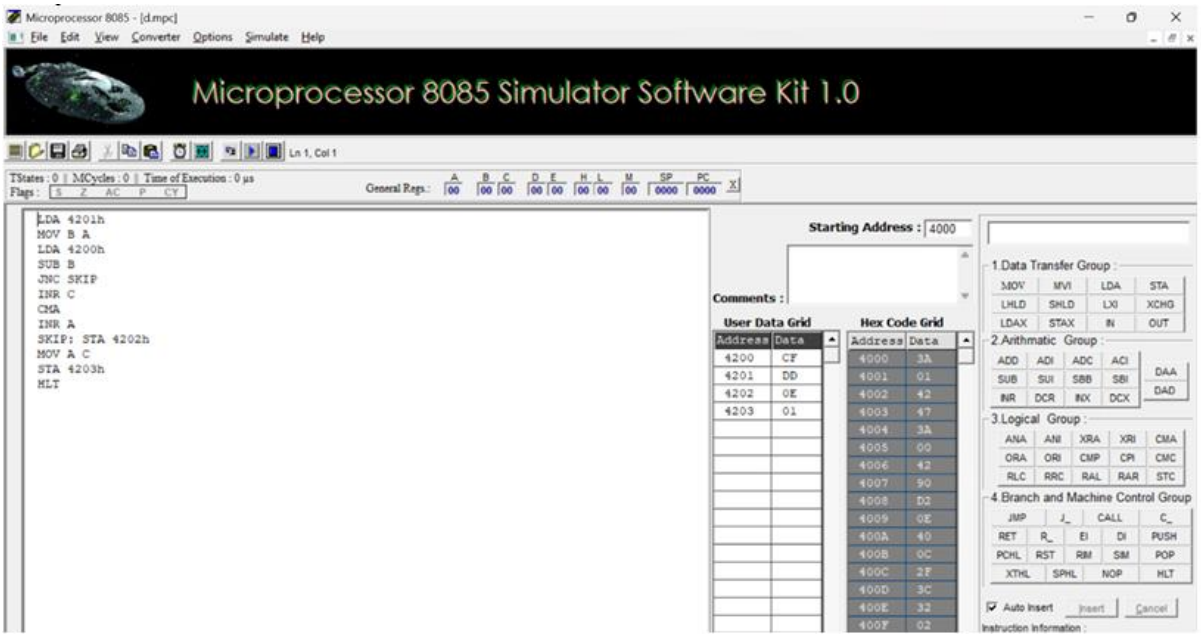
ALGORITHM:

1. Load the first number from memory into register B.
2. Load the second number into register C.
3. Initialize the accumulator to zero.
4. Add the first number repeatedly according to the second number.
5. Store the result in memory location 4202H.
6. Stop the program.

PROGRAM:

```
LDA 4200H
MOV B,A
LDA 4201H
MOV C,A
MVI A,00H
LOOP: ADD B
DCR C
JNZ LOOP
STA 4202H
HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program to multiply two 8-bit numbers using the 8085 microprocessor was written and executed successfully.